

Pest and Disease Management

Pest of Coconut

Disease Management

Pest of Coconut

1. Rhinoceros Beetle: *Oryctes rhinoceros*
2. Coconut Eriophyid mite: *Aceria guerreronis*
3. Red palm weevil: *Rhynchophorus ferrugineus*
4. Leaf eating caterpillar (or) Black headed caterpillar: *Opisina arenosella*
5. Slug caterpillar: *Parasa lepida* and *Contheyla rotunda*
6. Coconut skipper: *Gangara thyrsi*; *Suastus gremius*
7. Coreid bug: *Paradasynus rostratus*
8. Bag worm: *Manatha albipes*
9. White grub: *Leucopholis coneophora*
10. Termite: *Odontotermes obesus*
11. Lacewing bug: *Stephanitis typicus*
12. Scale insect: *Aspidiotus destructor*
13. Mealy bug: *Pseudococcus longispinus*
14. Palm Civet: *Vivera zibatha*
15. Rat: *Rattus rattus wroughtoni*
16. Nut Borer: *Cyclodes omma*
17. Nematodes

1. RHINOCEROS BEETLE: *Oryctes rhinoceros*

Pest population occurs round the year but population maximum during June – Sep coinciding with the onset of monsoon.



Rhinoceros Beetle



Rhinoceros Grub

Symptoms of Damage

- The adult beetle bores into the unopened fronds and spathes. Damage by the pest leads to 10 to 15% loss in yield.
- The attacked frond when fully opened shows characteristic triangular cuts.
- Central spindle appears cut or toppled
- Fully opened fronds showing characteristic diamond shaped cuttings
- Holes with chewed fibre sticking out at the base of central spindle.



Holes with chewed fibre sticking out in central spindle



Triangular cuts on leaves

Identification of the Pest

Egg: Oval creamy white egg in manure pits or decaying vegetable matter at a depth of 5 to 15 cm. Egg periods is 8 to 18 days. Female laid 140 to 150 eggs.

Grub: Grub is stout, sluggish, white “C”-shaped with pale brown head and found at a depth of 5 to 30 cm.

Pupa: Grub pupates in earthen cells at a depth of 0.3 to 1 m

Adult: Adult beetle is stout, brownish black or black and has a long horn projecting dorsally from the head in male. Horn is short in female.

Management

(i) Cultural Method:

- Remove and burn all dead coconut trees in the garden (which are likely to serve as breeding ground) to maintain good sanitation.
- Collect and destroy the various bio-stages of the beetle from the manure pits (breeding ground of the pest) whenever manure is lifted from the pits.

(ii) Mechanical Method:

- During peak period of population build up, the adult beetle may be extracted from the palm crown using GI hooks.
- Set up light traps following the first rains in summer and monsoon period to attract and kill the adult beetles.

(iii) Chemical Method:

Rhinoceros Beetle management - Chemical Method - Video

- The topmost three leaf axils around the spindle may be filled with any of the following mixtures as a prophylactic measure:
(a) Sevidol 8G 25 g + fine sand 200 g, which is to be done thrice in a year in April-May, September-October and December-January.
(b) For seedlings, apply Naphthalene balls 10.5 g (approx. three to four balls) covered with fine sand, once in 45 days.
- Place phorate 10 G 5 g in perforated sachets in two inner most leaf axils for 2 times at 6 months intervals.
- Treat manure pits and other possible breeding sites with 0.01% carbaryl (50 % WP) on w/w basis. Treatment will have to be repeated every three months.

(iv) Trap Method:



Rhinoceros Beetle



Trapped Rhinoceros Beetle

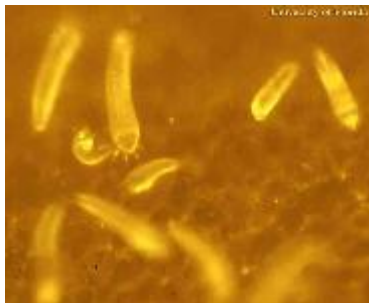
- Set up Rhino lure pheromone trap @ 5 traps/ha to trap and kill the beetles. The dispenser may be hanged in a plastic bucket having 2 liter of insecticide solution once in a week. Trapped beetles can be disposed off.

(v) Biological Method:

- Application of green muscardine fungus, *Metarrhizium anisopliae* @ 5 x 10¹¹ spores / m³ - spray 250ml Metarrizhium culture + 750ml water in manure pits to check the perpetuation of the pest.
- Field release of *Baculovirus oryctes* inoculated adult rhinoceros beetle @ 15 beetles/ha reduces the leaf and crown damage caused by this beetle.
- Soak castor cake at 1 kg in 5 liter of water in small mud pots and keep them in the coconut gardens to attract and kill the adults.
- Apply mixture of either neem seed powder + sand (1:2) @150 g per palm or neem seed kernel powder + sand (1:2) @150 g per palm in the base of the 3 inner most leaves in the crown.

[↑Top of page](#)

2. COCONUT ERIOPHYID MITE: *Aceria guerreronis*



Colony of coconut mite



Eriophyid mite under microscope

Symptoms of Damage

- The earliest symptom on 2-3 month old buttons is pale yellow triangular patches seen below the perianth.
- Later, these patches become brown. Severely affected buttons may fall. As the buttons grow, brown patches lead to black necrotic lesions with longitudinal fissures on the husk.
- Oozing of the gummy exudation from the affected surface of the nuts.

- Uneven growth results in distortion and stunting of nuts leading to reduction in copra yield. In severe cases, the nuts are malformed with cracks and hardened husk.



Yellow patches on leaves



Brown colour patches on nuts

Identification of the Pest

Mite are usually found under the bracts of fertilized female flowers and do not infest the unfertilized flowers. This mite is very minute in size measuring 200 – 250 micron in length and 36 – 52 micron in width with two pairs of legs. Nymph and Adult is pale in colour with elongate body and worm like appearance. The life cycle of this mite, which consists of egg, two larval instars and an adult stage, is completed in 7 -10 days.

Management

(i) Cultural Method:

- Collect and destroy all the fallen buttons of the affected palm.
- Grow intercrop (sun hemp, four crops/year) and shelter belt with casuarina all round the coconut garden to check further entry.
- Providing adequate irrigation.
- Apply urea 1.3 kg, super phosphate 2.0kg and muriate of potash * 3.5 kg/palm/year.

** Increased quantity is recommended to increase the plant resistance to the mite.*

- Soil application of micro nutrients like, Borax 50 g + gypsum 1.0kg + Manganese sulphate 0.5 kg/palm/ year TNAU Micronutrient solution 200 ml/tree.
-

Eriophyid Mite Management - Chemical Method - Video

(ii) Chemical Method:

i. Root Feeding:

- Monocrotophos 36 WSC @ 15ml or triazophos 40 EC @15 ml or carbosulfan 25EC @ 15 ml / 15 ml of water
- After root feeding, next harvest should be done 45 days later.
- TNAU - Agro biocide - 30 ml/tree - (60 days after Carbosulfan root feeding).
Note: Pluck nuts before root feeding

ii. Spot application of ecofriendly Botanicals

Round 1: Azadirachtin 1% (5 ml in one lit. of water)

Round 2: Neem oil + Teepol (30 ml in one lit. of water)

- Triazophos 40 EC 5 ml/lit or monocrotophos 36 WSC @ 2 ml / lit or carbosulfan 25 EC 2 ml/ lit in alternation with neem azal 1% 5ml/lit as spot application
- Neem cake application @ 5 kg per palm per year
- Preparation of neem oil + garlic emulsion (2%):
 - To prepare 10 litres of 2% neem oil + garlic emulsion, 200 ml neem oil, 200 g garlic and 50 g ordinary bar soap are required.
 - Slice the bar soap and dissolve in 500 ml lukewarm water.
 - Grind 200 g of garlic and take the extract in 300 ml of water.
 - Pour the 500 ml soap solution in 200 ml neem oil slowly and stir vigorously to get a good emulsion.
 - Mix the garlic extract in the neem oil + soap emulsion.
 - Dilute this 1 litre stock solution by adding 9 litres of water to get 10 litres of 2 % neem oil + garlic emulsion.

Method of Application

- The botanicals should be applied in the sequence indicated above at 45 days interval using a one litre hand sprayer. Rocker or Pedal sprayer can be used for spraying small trees.
- The spray should be applied at the crown region by a climber covering only the top six bunches during non rainy season.

Precautions and Safety Measures

- Spraying should be avoided during windy season to prevent contamination.
- At the time of spraying, protective mask and clothing should be used.
- Wash face and hands cleanly with soap after spraying.

Biological Method:

Entomofungal pathogen *Hirsutella thompsonii* and *Verticillium lecanii* are reported to be promising in managing the mites. Both the pathogens are mass multiplied by a commercial firm and sold in the market.

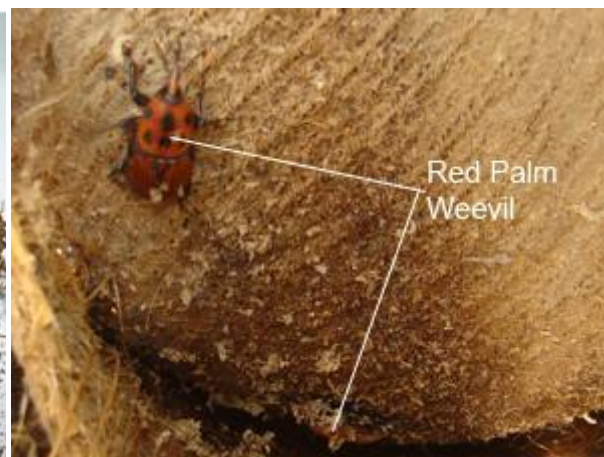
[↑Top of page](#)

3. RED PALM WEEVIL: *Rhynchophorus ferrugineus*

It occurs round the year but becomes severe after monsoon



Red palm weevil pupa with case



Red palm weevil

Symptoms of Damage

Red palm weevil is one of the most destructive pests of coconut, oil palms and ornamental palms.



Holes on the trunk



Chewed up fibrous matter from the hole

- The hole can be seen on the stem with chewed up fibres protruding out.
- Many times reddish brown liquid can be seen oozing out from the hole.
- The grubs cause damage inside the stem or crown by feeding on soft tissues and often cause severe damage especially when a large number of them bore into the soft, growing parts. In case of severe infestation the inside portion of trunk is completely eaten and become full of rotting fibres.
- In case of young palms the top withers while in older palms the top portion of trunk bends and ultimately breaks at the bend (wilting).
- Sometimes the gnawing sound produced by the feeding grubs inside will also be audible.
- In the advanced stage of infestation yellowing of the inner whorl of leaves occur. The crowns falls down or dry up later when palm is dead.

Identification of the Pest

Egg: Oval and creamy white in colour. Eggs laid in scooped out small cavities, wounds and other cut injuries on the trunk

Grub: Light yellowish grub without legs. Stout, fleshy and apodous with a conical body bulged in middle and tapering towards the end

Pupa: The full grown larva pupates inside the stem and fibrous cocoon made out of fibrous strands

Adult: Reddish brown weevil has six dark spots on thorax. Male has conspicuous long snout has a tuft of hairs.

Management

(i) Cultural Method:

- Remove and burn all wilting or damaged palms in coconut gardens to prevent further perpetuation of the pest.
- Avoid the cutting of green leaves. If needed, they should be cut about 120 cm away from the stem in order to prevent successful inward movement of the grubs through the cut end.

(ii) Trap Method:

Red Palm Weevil - Trap Method - Video

- Coconut log traps: Setting up of attractant traps (mud pots) containing sugarcane molasses 2½ kg or toddy 2½ litres (or pineapple or sugarcane activated with yeast or molasses) + acetic acid 5 ml + yeast 5 g + longitudinally split tender coconut stem/logs of green petiole of leaves of 30 numbers in one acre to trap adult red palm weevils in large numbers. Incorporate any of the insecticide to each trap to kill the weevils trapped.
- Install pheromone trap @ one trap per ha

Step 1: Specialized buckets with 3 or 4 holes are made, the bucket is wound with coconut fibre/ jute sack, so that the pests can enter.

Step 2: The lure (Ferrolure +) is suspended inside the bucket and one lit of water is added along with 100g pineapple/ sugarcane, 2g yeast and 2g Carbaryl in the bucket.

Step 3: The bait buckets are placed at sites in the farm, where infestation is seen most.

Step 4: After a week the water is checked for the catch & re filled to prevent mosquitoes from breeding.

Chemical Method:

Red Palm Weevil - Chemical Method - Video

- In attacked palms, observe for the bore- holes and seal them except the top most one. Through the top most hole, pour 1% carbaryl (20gm/lit) or 0.2% trichlorphon @ one litre per palm, using a funnel. Then plug this hole also. If needed repeat after one week.
- When the pest infestation is through the crown, clean the crown and slowly pour the insecticidal suspension.
- In case of entry of weevil through the trunk, the hole in trunk may be plugged with cement / tar.

- A slanting hole is made with the aid of an auger and the insecticide solution is poured with funnel.
- Fill the crown and the axils of top most three leaves with a mixture of fine sand and neem seed powder or neem seed kernel powder (2:1) once in three months to prevent the attack of rhinoceros beetle damage in which the red palm weevil lays eggs.

Root Feeding

- Select a fresh and live root
- Cut sharply at an angle and insert the root in the insecticidal solution containing monocrotophos 36 WSC 10 ml + water 10 ml in a 7 x 10 cm polythene bag.
- Secure the bag tightly to the root with a cotton thread.
- Twenty four hours later, check whether there is absorption.
- If there is no absorption select another root.
- These methods should not be resorted to as a routine practice and it is suggested only for cases of severe epidemic outbreak of the pest and when the survival of the tree is threatened.

[↑Top of page](#)

4. LEAF EATING CATERPILLAR / BLACK HEADED CATERPILLAR: *Opisina arenosella*

This causes severe damage to palms in coastal and back water areas and in certain internal packets of peninsular in India. The pest occurs round the year with the spike in population during summer (Mar-May).



Black Headed Caterpillar - Adult



Black Headed Caterpillar - Larva

Symptoms of Damage



Galleries of silk and frass seen on underside of leaves

- The coconut trees of all ages are attacked.
- Dried up patches on leaflets of the lower leaves, only three or four youngest leaves at the center of the crown remain green.
- Galleries of silk and frass on under side of leaflets.
- In case of severe infestation the whole plantations present a scorched appearance.
-

Black Headed Caterpillar Symptoms - Video

Identification of the Pest

Larva: Caterpillar is greenish brown with dark brown head and prothorax, and a reddish mesothorax. It has brown stripes on the body.

Pupa: It pupates inside the web itself in a thin silken cocoon

Adult Moth: Greyish white in colour

Female: with long antenna and three faint spots on the forewings

Male: with fringed hairs in hind wings in apical and anal margin.

Management

Cultural Method:

Black Headed Caterpillar Management (Mechanical/Cultural Method) - Video

- As a prophylactic measure, the first affected leaves may be cut and burnt during the beginning of the summer season.

Biological Method:

Black Headed Caterpillar Management (Biological Method) - Video

- Release the larval (*Bethylid*, *Braconid* and *Ichneumonid*) and pupal (*Eulophid*) on (chalcid) parasitoids and predators periodically from January, to check the build up of the pest during summer.
- Among the larval parasitoids, the Bethylid, *Goniozus nephantidis*, (*Elasmus nephantidis* (brown species) and *Brachymeria nosatoi* – KAU) is the most effective in controlling the pest. The optimum level of release is 1:8 of host-parasitoid ratio. The parasitoid should be released @ 3000/ha under the coconut trees when the pest is in the 2nd or 3rd instar larval stage. Parasitoid release trap may be used to release the parasitoid at the site of feeding. Parasitoids should not be released in the crown region since they will be killed by predators like spiders and reduviid bugs. This should be released only three weeks after chemical spray.

Chemical Method:

- When infestation is very severe in young palms and if the biocontrol is not likely to be effective, spray the undersurface of the fronds with dichlorvos 0.02% (Dichlorovos 100EC), malathion 50 EC 0.05% (1 ml/lit), quinalphos 0.05% or phosalone 0.05%.

Note: Application of the insecticides should be followed by liberation of larval and pupal parasites from the 21st day.

Root feeding

- Select a fresh and live root
- Cut sharply at an angle and insert the root in the insecticidal solution containing Monocrotophos 36 WSC 10 ml + water 10 ml in a 7 x 10 cm polythene bag.
- Secure the bag tightly to the root with a cotton thread.
- Twenty four hours later, check whether there is absorption.
- If there is no absorption select another root.
- These methods should not be resorted to as a routine practice and it is suggested only for cases of severe epidemic outbreak of the pest and when the survival of the tree is threatened.

Stem injection

- A Slanting hole is drill one meter above the ground level and 10 ml of Monocrotophos 36WSC is taken in a syringe and the needle is guided to the hole. After the insecticide has been absorbed the hole has to be plugged with clay mixed with copper oxy chloride.

[↑Top of page](#)

5. SLUG CATERPILLAR: *Parasa lepida* and *Contheyla rotunda*

Symptoms of Damage

The larvae is the destructive stage if the pest. The young larvae feed on the lower epidermis of the leaf. As they mature, the whole leaf blade is eaten leaving the midribs. In heavy infestation, the larvae may defoliate the palm.



Slug Caterpillar - Symptoms

Identification of the Pest 1

Parasa lepida

Egg: Flat shiny eggs on the under surface of leaves

Larva: Greenish body with white lines and four rows of spiny scoli tipped red or black, which cause irritation and pain.

Pupa: It pupates in a compact elliptical chocolate brown shell like cocoon, which is convex above and flat ventrally as stems. Cocoons are covered with irritating spines and hairs

Adult: Adult moth has green wings with prominent dark patch at the base of each forewing.

Management

Chemical Method:

1. Collect and destroy the immature stages of the insects by conducting study (or neem campaign) wherever possible and spray carbaryl 50 WP 2 gm/lit .

2. Root feeding with monocrotophos 36 WSC @ 10 ml + 10 ml water at 45 days interval for 3 times for control of leaf caterpillar.
3. Spray dichlorvos 76 WSC 2 ml / lit.

Mechanical Method:

1. Set up light traps to trap and collect adult moths. About 5 light traps may be installed per hectare.

Identification of the Pest 2:

Pest Name: **C.rotunda**

Larva- Black or grey dorsally and dorso laterally.

Adult - Small grayish brown moth. Forewings have slight dark in colour and series of black points. Hind wings are slightly darker.

Management

Chemical Method: Spray any one of the following

- Dichlorvos 76 WSC 2 ml/lit
- *Bacillus thuringiensis* 2 g/lit,
- Triazophos 40 EC 5 ml
- Methyl demeton 25 EC 4 ml/lit
- Root feeding with monocrotophos 15 ml + 15 ml of water

[↑Top of page](#)

6. COCONUT SKIPPER: *Gangara thyrsis* ; *Suastus gremius*

Symptoms of Damage



Leaflets tips are rolled

Skipper larvae

- One half of the leaflets are cut and rolled into a case
- The rolled leaflets are dried

Identification of the Pest



Gangara thyrsis pupa



Coconut skipper larvae

Larva: Pale green with reddish markings. Body concealed in a covering of white waxy markings.

Adult: Butterfly-brownish in colour. Chocolate brown wing with yellow spots.

Management

Chemical Method:

1. Collect and destroy the immature stages of the insects by conducting study (or neem campaign) wherever possible and spray carbaryl 50 WP 2 gm/lit.
2. Root feeding with monocrotophos 36 WSC @ 10 ml + 10 ml water at 45 days interval for 3 times for control of leaf caterpillar.
3. Spray dichlorvos 76 WSC 2 ml / lit.

Mechanical Method: Set up light traps to trap and collect adult moths.

[↑Top of page](#)

7. COREID BUG: *Paradasynus rostratus*

It occurs in coastal areas and high ranges in Kerala (Trivandrum, Wynad, Kasaragod).

Symptoms of Damage



Immature nut fall



Crevices on the husk below the perianth

The adults and nymphs feed by desapping the contents on button and developing nuts below the perianth region. The attacked buttons become deformed with characteristic crevices on the husk below the perianth with gum exudations and the tender nuts become barren. Severe damage leads to nut fall and malformation of mature nuts.

Identification of Pest

The adults are brown colored measuring 2 cm in size. The life cycle is completed in a month's time.

Management

Chemical Method:

Coreid Bug Management (Chemical Medhod) - Video

1. Collect and destroy the immature stages of the insects by conducting study (or neem campaign) wherever possible and spray carbaryl 50 WP 2 gm/lit.
2. Root feeding with monocrotophos 36 WSC @ 10 ml + 10 ml water at 45 days interval for 3 times for control of leaf caterpillar.
3. Apply 0.1% carbaryl on the newly opened inflorescence after the receptive phase of the female flowers and spray the entire crown excluding the leaves and older bunches (at 45 days interval). Destruction of pollinating insects can be avoided if spraying is done in afternoon hours.

4. Spray dichlorvos 76 WSC 2 ml / lit.

Mechanical Method: Set up light traps to trap and collect adult moths.

[↑Top of page](#)

8. BAG WORM: *Manatha albipes*

Symptoms of Damage

Small and irregular hole on the leaves. Silken spin stick strand bags below the leaf.



Symptoms of Bag Worm

Bag Worm Larva

Identification of the Pest

Larva: Inside the conical bags constructed with silk and thread

Management

Chemical Method:

1. Collect and destroy the immature stages of the wherever possible and spray carbary 50 WP 2 gm/lit.
2. Root feeding with monocrotophos 36 WSC @ 10 ml + 10 ml water at 45 days interval for 3 times for control of leaf caterpillar.
3. Spray dichlorvas 76 WSC 2 ml / lit.

Mechanical Method: Set up light traps to trap and collect adult moths.

[↑Top of page](#)

9. WHITE GRUB/ COCKCHAFER BEETLE: *Leucopholis coneophora*

These cream colored grubs are found in abundance in sandy loam tracts of Kerala and Karnataka, especially in Sep – Oct and the beetles emerge in May-June.

Symptoms of Damage

1. In nursery the grubs feed the tender roots and tunnel into the bole of the collar region resulting in drying and yellowing
2. In older plantations infestations results in yellowing of leaves, premature nut shedding, delayed flowering, retardation in growth and yield decline.
3. White grubs are exposed when base of tree dug
4. tuber crops grown as intercrops are also affected.

Identification of the Pest

Adult: Brown coloured beetle with striated wings not covering the abdomen fully

Egg: Oval in shape and creamy white in color when freshly laid. Prior to hatching they turn to dirty white.

Larvae: Curved, fleshy and wrinkled. They are creamy white in color with brown head.

Management

Cultural method:

- Repeated ploughing once a week for 4 to 5 times after first rains in summer reduce the pest population by exposing the pest to predation by birds and other animals.
- Plant neem twigs with leaves in coconut gardens after rain to attract and kill adult beetles
- Plough or dig the infested soil synchronizing with pre-monsoon showers.

Physical Method:

- Collect and destroy the adult beetles attracted to trees like neem, *Ailanthus* and *Acacia* on the receipt of monsoon showers (in the evening).

Mechanical Method:

- Set up light trap @ 1 / ha or bonfire

Chemical Method:

Soil application - Malathion 5 D 25 kg/ ha at the time of planting (Treat the soil with phorate 10G @ 100 g/palm or drench with chlorpyrifos 0.04% suspension. The treatment should be

given twice, first during April-May after the receipt of pre-monsoon showers and second during the month of September - Recommended by KAU).

[↑Top of page](#)

10. TERMITE: *Odontotermes obesus*

Symptoms of Damage



Termite Symptoms - Soil runway made at the base of the trunk

- Termites are likely to cause damage to transplanted seedlings particularly in the earlier stage. (wilting of seedlings)
- Base of trunks plastered with runways made of soil and fibers

Identification of the Pest

Adults: Cream coloured, tiny insects resembling ants with dark coloured head

Management

Cultural Method:

1. Locate termite mounds in or near the coconut nursery or garden and destroy.
2. Adoption of field sanitation by disposal of organic matter in nursery soil and covering germinating nuts with a layer of river sand.

Chemical Method:



Termite Control - Apply calcium at base of the trunk

- Spray Copper Sulphate 1% or Cashew Nut Shell oil 80% or spray Chlorpyrifos @ 3ml/lit of water, Neem Oil 5% or NSKE 20% to preserve plaited coconut leaves from the termite attack.
- Apply calcium at the base of the trunk for control of termite attack.
- Swabbing with neem oil 5% once on the base and upto 2 m height of the trunk for effective control.

[↑Top of page](#)

11. LACEWING BUG: *Stephanitis typicus*

Symptoms of Damage

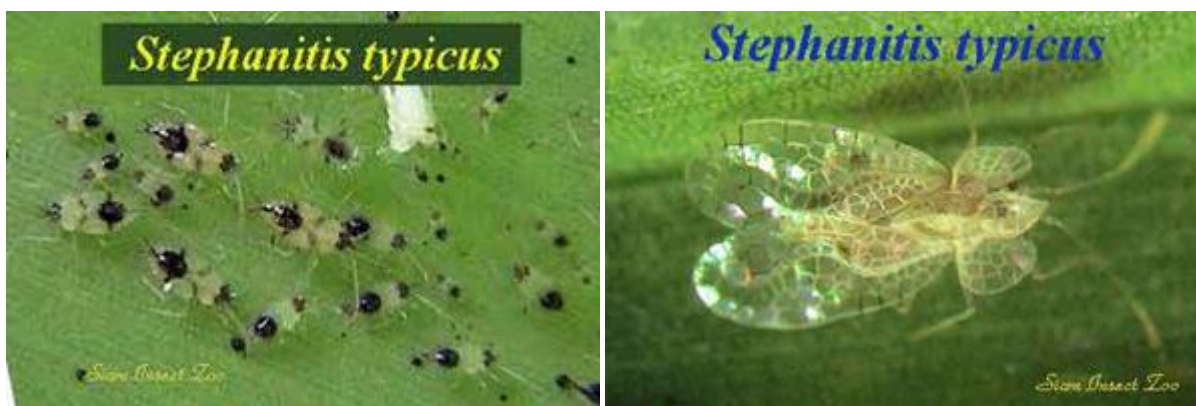


Lacewing Bug - Symptoms

- White spots on the upper surface of the leaves

Lace bug sucks sap from coconut foliage; it acts as a vector in transmission of Phytoplasma from root (wilt) affected palms to healthy palm.

Identification of the Pest



Lacewing Bug - Identification

Nymph: White with dark patches.

Adults: White coloured adults with netted venation on wing.

Management

Cultural Method: Remove leaflets harboring these insects and destroy them.

Chemical Method: Spray any one of the following;

- Malathion 50 EC @ 2 ml/lit
- Dimethoate 30 EC @1 ml/lit
- Methyl demeton 25 EC @1 ml/lit
- Phosphamidon 40 SL @1.25 ml/lit
- Monocrotophos 36 WSC @1 ml/lit
- Methomyl 25 EC @2 ml/lit
- 3% Neem oil

[↑Top of page](#)

12. SCALE INSECT: *Aspidiotus destructor*

Symptoms of Damage: It occurs more in summer. Scale insects affects on leaves and nuts of coconut palms. A severe infestation, scale forms a continuous crust over flower spikes, young nuts and lower surface of leaves.



Symptoms of Scale Insect

On leaves, *A. destructor* causes scales with yellow spots developing where the crawlers have settled and grown into adults. Entire leaves may turn yellow to brown and fall. Sooty mould may develop. The bright yellow colour of affected coconut palms is clearly visible from a great distance. In extreme cases, the leaves dry out, entire fronds drop off and the crown dies. Heavy infestation results in stunting of new leaves, reduction of crop yield or complete crop failure. Infested coconut fronds exhibit yellow areas on the upper surface, formed by

numerous yellow spots each marking the position of the coconut scale on the under surface.

Identification of the Pest



Scale colonies



Scale insect on leaf

Egg: hatching the young scales on the undersides of the leaves.

Nymph: Covered with circular waxy secretion

Adult: The scale is bright yellow and round or reddish (female) and oval (male) covered with semitransparent grayish white flat scale. Females are always wingless and remain under their scale their entire life. Adult males have one pair of membranous wings, move about actively in search of females and do not feed during adult stage.

Management

Chemical Method:

1. Spray Fish Oil Rosin Soap (FORS) 2.5% or spray Fenthion / Malathion. A second round is given after 20 days.
2. Pluck mature nuts and spray monocrotophos 36 WSC @ 2 ml/ha. (or 1ml/ha)
3. Do not harvest nuts for 45 days after spraying.

Biological Method:

- Release of predatory Coccinellids, *Chilocorus nigritus* is found to be effective.

[↑Top of page](#)

13. MEALY BUG: *Pseudococcus longispinus*

Symptoms of Damage

Mealy bugs colonize on all tender plant parts like bases of spear leaf, spadix and inflorescence and beneath the perianth of the nut.



Mealy Bug - Symptoms & Damage

Mealy bugs infest the unopened heartleaf and inflorescence. It feeds plant sap. Leaves are yellowing and dry up. As a result, the leaves become highly stunted, suppressed, deformed and present a crinkled appearance. It is often confused with the leaf rot symptoms. The affected inflorescences are malformed and do not open. Even if they open, they do not bear nuts.

Button mealy bugs colonize under the perianth lobes of tender nuts. Infested nuts harbouring gravid mealy bugs remain on the spadix, which serve as inoculum for further spread.

Identification of the pest

Nymph Nymphs are flat, oval and yellow. Older nymphs of some species are covered with fluffy, white wax.

Adult: The males are yellowish in color whereas the females are longer and narrower and white in color.

Management

Cultural Method: Remove leaflets harbouring these insects and destroy them.

Chemical Method: Spray any one of the following;

- Malathion 50 EC @ 2 ml/lit
- Dimethoate 30 EC @1 ml/lit

- Methyl demeton 25 EC @1 ml/lt
- Phosphamidon 40 SL @1.25 ml/lt
- Monocrotophos 36 WSC @1 ml/lt (in seedling stage 0.05%)
- Methomyl 25 EC @2 ml/lit
- 3% Neem oil

[↑Top of page](#)

14. PALM CIVET *Viverra zibatha*

Indian civets have large bodies that are gray or brown in color. Body length is about 34 inches with a tail length of 13 inches. They have black spots on the body as well as black and white stripes on the sides of the neck. In most cases there are two white stripes and three black stripes. The tail has a number of black rings around it. Limbs are black and the forefeet contain lobes of skin on the third and fourth digit that protect the retractile claws. Males are slightly larger than females.



Palm Civet

Symptoms of Palm Civet

Management

Chemical Method: Poison baiting with ripe banana fruit sandwiched with 0.5 g carbofuran 3 G granules.

[↑Top of page](#)

15. RAT: *Rattus rattus wroughtoni*

Symptoms of Damage

Rats damage tender nuts by forming characteristic holes. Shed nuts can be seen at the base of the palm.



Symptoms of Rat Damage

Management

Mechanical Method:

1. Tree banding with inverted iron cones (Entry of rats on to the trunk can be prevented by fixing mechanical barriers upto 2m height from ground level using 40cm sized G.I. sheets) or Prosopis thorns.
2. Use rat traps.



Rat Trap - Mechanical Medhod

Chemical Method:

Rat Management (Chemical Method) - Video

- Baiting with bromodialone 0.005% at 10 g/tree at crown region twice at an interval of 12 days.
- Fumigate the hiding places using Aluminium phosphide tablets.
- Poison baits prepared by mixing 95 parts raw rice, three parts coconut oil and two parts zinc phosphide are placed in active burrows.

[↑Top of page](#)

16. NUT BORER: *Cyclodes ommata*

Symptom of Damage

The caterpillar bores into the developing buttons at perianth portion causing nut drop.

Nut borer - Video

Management

- Hand picking and destroying
- Spray 0.1% Carbaryl

[↑Top of page](#)

17. NEMATODES

Radopholus similis: Among nematodes, **Radopholus similis** or the **burrowing nematode** damage the roots of coconut. Burrowing nematode populations survive under field conditions for six months in moist soil (27 to 36°C) and one month in dry soil (29 to 39°C), whereas it survives for 15 months in moist soil (25.5 to 28.5°C) and 3 months in dry soil (27 to 31°C) under greenhouse conditions. The nematode survives in roots of stumps of felled coconut palms up to six months.

In coconut plantations in Kerala, seedlings are raised by sowing seednuts in the interspaces of coconut palms. The burrowing nematode is widely prevalent in most of the nurseries in Kerala and Tamil Nadu. One – year – old coconut seedlings raised in such infested nurseries harbor large populations of the burrowing nematode within roots both internal and external

of the husks. These infested seedlings when distributed to distant places through government as well as private agencies for planting help in the dissemination of the nematode.

Symptoms of Damage

The nematode infested coconut palms exhibit general decline, yellowing, button shedding, and reduction in leaf size. The symptoms on roots are more specific. Elongated orange colour lesions are seen on tender and semi hard roots. Consequent to nematode parasitization and multiplication these lesions enlarge and coalesce to cause extensive rotting of roots. Tender roots on heavy infestation become spongy in texture.

Management

Biological Method:

1. Application of cowdung, farmyard manure, oil cakes and green manure to the basins. *Crotolaria juncea* may be cultivated in the basin and interspaces and used as green manure.
2. Incorporate leaves and tender stem of *Crotolaria juncea*, *Pueraria javanica* and *Glyricidia maculate* into the soil in Sep-Oct.

Chemical Method:

1. This can be controlled with application of phorate 10G @ 100 g/palm/twice a year. That is during May-June and Sep- Oct.

Cultural Method:

- Use of less susceptible, tolerant cultivars or hybrids of coconut and intercrops in infested areas.
- Avoid use of banana as a shade crops in coconut nurseries

[↑Top of page](#)

Disease Management

[1. Bud Rot: *Phytophthora palmivora*](#)

[2. Tanjore Wilt \(or\) Basal stem end rot: *Ganoderma lucidum*](#)

[3. Root \(wilt\) Disease: unknown etiology](#)

[4. Leaf rot disease](#)

[5. Grey Leaf Spot: *Pestalotiopsis palmarum*](#)

[6. Stem bleeding disease](#)

[7. Leaf blight: *Lasiodiplodia theobromae*](#)

[8. Crown chocking](#)

1. BUD ROT: *Phytophthora palmivora*

Symptoms of Damage

Palms of all age are liable to be attacked but normally young palms are more susceptible, particularly during monsoon when the temperature is low and humidity is very high. In seedlings, the spear leaf turns pale and comes off with a gentle pull.



BUD ROT: Symptoms of Damage

- The earlier symptom is the yellowing of one or two younger leaves. Black spots appear on spindle leaves. Basal tissues of the leaf rots quickly and can be easily separated from the crown.
- In the later stages the spindle withers and drops down.
- The tender leaf base and soft tissues of the crown will rot into a slimy mass of decayed material emitting foul smell.
- Ultimately the entire crown falls down and the palm dies



BUD ROT: Symptoms in Adult Palm

In adult palms, the first visible symptom is the colour change of the spear, which becomes pale and breaks at the base and hangs down. The rotting slowly progresses downwards, finally affecting the meristem and killing the palms. This is accompanied by drooping of successive leaves. Even then, nuts that are retained on the palm may grow to maturity.

Management

Chemical Method:

- Remove all the affected tissue of the crown region and crown drenching with Copper oxychloride 0.25%. (Apply Bordeaux paste and protect it from rain till normal shoot emerges. (Dissolve 100 gm of copper sulphate and 100 gm of quick lime each in 500ml. water separately and mix to form 1 litre of Bordeaux paste).
- Spray 0.25% Copper oxychloride on the crown of the neighbouring palms as a prophylactic measure before the onset of monsoon. (Spray 1% Bordeaux mixture on spindle leaves and crown of disease affected as well as neighbouring palms, as a prophylactic measure. Palms that are sensitive (Dwarf palms) to copper containing fungicides can be protected by mancozeb. Small, perforated sachets containing 2 g of mancozeb may be tied to the top of leaf axil. When it rains, a small quantity of the fungicide is released from the sachets to the leaf base, thus protecting the palm.
- The infected tissues from the crown region should be removed and dressed with Bordeaux paste sprayed with 1% Bordeaux mixture as pre-monsoon spray (May and September).
- Leaf axil filling with Sevidol 8G, 25 g mixed with 200g sand is recommended to red palm weevil infestation of affected palms.
- Spray with Copper oxychloride 0.25% after the onset of Monsoon.

Cultural Method:

- Provide adequate drainage in gardens.
- Adopt proper spacing and avoid over crowding in bud rot prone gardens.

[Top of page](#) | [Disease Management](#)

2. TANJORE WILT / BASAL STEM END ROT; Ganoderma Wilt

This disease first appeared in the coastal areas of Thanjavur district of TamilNadu after cyclones of 1952 and 1955 and hence the names Thanjavur wilt. In Tamil Nadu it is

prevalent in Thanjavur and Chengulpet districts. In Karnataka it is found in Maidan areas. This disease is of recent occurrence in many parts of Kerala, especially in the districts of Palakkad, Malappuram, Thrissur, Kollam, Thiruvananthapuram and Wayanad.

Causal Organism: *Ganoderma lucidem* and *Ganoderma applanatum*.

Symptoms of Damage



Stem bleeding

*Basal stem rot -
bracket formation*

Drooping of leaves

Big fructification

TANJORE WILT / BASAL STEM END ROT: Symptoms of damage

- Initial symptoms of Thanjavur wilt (Ganoderma wilt) start with withering, yellowing and drooping of the outer whorl of leaves.
- This is followed by exudation of reddish brown liquid through cracks at the base of the trunk and oozing spread upward. The tissues on the bleeding spots are soft to touch.
- Decaying of tissues at bleeding point and rotting of the basal portion of the stem
- The bark turns brittle and often gets peeled off in flakes, leaving open cracks and crevices. The internal tissues are discoloured and disintegrated, emitting a bad smell.
- Bracket formation at the base of the trunk. Ganoderma appears at the base of the trunk. Ultimately the palm dies off

Management

Cultural Method:

- Remove and destroy all affected palms.
- Green manure crops must be raised and ploughed in-situ at the time of flowering.

Biological Method:

- Apply *Pseudomonas fluorescens* (Pf1) @ 200g/palm + *Trichoderma viride* @ 200g/palm/year.
- Apply 200g Phosphobacter and 200 g Azotobactor mixed with 50kg of FYM/palm.
- Apply FYM 50kg + neem cake 5 Kg once in 6 months along with fertilizers.

Chemical Method:

- Isolation of trench around the tree, 4 feet away from the base of the trunk. Apply Sulphur dust inside the trench.
- The bleeding patches in the stem may be chiseled and protected with tridemorph (5% calxin) and subsequently with hot coal tar.
- Aureofungin-sol 2 g +1 g Copper Sulphate in 100ml water or 2 ml of Tridemorph in 100 ml water applied as root feeding. (The active absorbing root of pencil thickness must be selected and a slanting and a slanting cut is made. The solution to be taken in a polythene bag or bottle and the cut end of the root should be dipped in the solution).
- Trunk injection / root feeding with Calixin 3 ml/tree.
- Forty litres of 1% Bordeaux mixture should be applied as soil drench around the trunk in a radius of 1.5m.

[Top of page](#) | [Disease Management](#)

3. ROOT WILT or KERALA WILT

Causal Organism: Phytoplasma. The disease is transmitted by lace bug *Stephanitis typica* and the plant hopper *Proutista moesta*.

Symptoms of Damage

- Tapering of terminal portion of the trunk.
- Reduction of leaf size
- Abnormal bending or Ribbing of leaf lets termed as flaccidity.
- Flowering is delayed and also yield is considerably reduced.



Flaccidity and bending of leaflets



Necrosis



Severely affected palm

ROOT WILT or KERALA WILT: Symptoms of damage

The characteristic symptom is the **flaccidity** of leaflets. This is the earliest visual symptom. In the beginning yellowing is restricted from the leaf tips to the middle of the leaves, necrosis of leaflets and deterioration and decay of root system are other salient features of the disease. The leaflets curve inwardly to produce ribbing so that the whole frond develops a cup like appearance. Abnormal shedding of buttons and immature nuts are also noticed.

Hot Spot Areas: Kottayam, Alapuzha, Pathanamthitta and Kollam districts of Kerala.

Coconut root (wilt) is a non-lethal debilitating disease and the affected palms survive for a long period giving a reasonably good yield. The root (wilt) affected palms are susceptible to diseases like leaf rot and pests like rhinoceros beetle and red palm weevil. So there is a chance of confusing the pests and disease symptom with the root (wilt) disease. Negligence on the management aspects aggravates the malady. Efficient management of palms suspected to be affected by coconut root (wilt) disease demands control of all pests and diseases and imparting natural resistance and health to the palms through proper manuring and agronomic practices.

Management

Cultural Method:

- Cut and remove disease advanced, uneconomical palms yielding less than 10 nuts per palm per year
- Grow green manure crops - cowpea, sunhemp (*Crotalaria juncea*), *Mimosa invisa*, *Calapagonium mucanoides*, *Pueraria phaseoloides* etc. may be sown in coconut basins during April-May and incorporated during September-October.

- Irrigate coconut palms with at least 250 litre water in a week.
- Adopt suitable inter/mixed cropping in coconut gardens.
- Provide adequate drainage facilities.
- Apply fertilizers for coconut palms in average management at the rate of 1.3 kg urea, 2.00 kg super phosphate and 3.5 kg potash (MOP) / palm / year in the form of urea, rock phosphate and muriate of potash, respectively.
- Magnesium may be supplied @ 500 g MgO per palm per year in (sandy soil) area. The cheapest source of MgO is magnesite (MgCO₃). The magnesium in magnesite is acid soluble. Hence it may be preferred in acid soils.
- To manage the insect vectors treat the top two leaf axils with insecticide preparation. This can be prepared by mixing phorate 10 G -20 g with 200 g sand or powered neem cake 250g mix with equal quantity of sand around the base of the spindle.

Water Management (Irrigated) of Root Wilt affected Garden:

Age of Palm	Basin	Drip	Perfo/Sprinkler (HDMSCS)
1-2 years old	25-30litres once in 2 days	10 lit per day, two emitters 50 cm from base	-
3-4 years old	75-80 litres once in 4 days	20lit per day, 3 emitters 75 cm from base	-
Adult Palm	200-250 litres once in 4 days	30-35 lit per day, 4 emitters 1 m away (in laterite); 6 emitters (literal sandy soil) Pit size for emitters – 25cm ³	Irrigate to a depth of 20 mm once in 4 days

Biological Method:

- In addition to the above, apply 50 kg FYM or green manure and 5 kg of neem cake / palm / year.
- Growing green manure crops like sunn hemp, sesbania, cowpea and calapagonium in the coconut basin and their incorporation *in situ* is beneficial as the practice reduces the intensity of the root (wilt) and increases the nut yield. The ideal green manure crops for the sandy and alluvial soils are cowpea and sesbania, respectively.
- Apply manures in 10 cm deep circular basins at a radius of 2 m from the bole of the palm.

4. LEAF ROT DISEASE

Symptoms of Damage: Leaf rot disease commonly occurs on coconut palms already affected by root wilt disease especially in 8 southern districts of Kerala, namely Thiruvananthpurma, Kollam, Alappuzha, Pathanamthitta, Kottayam, Ernakulam, Idukki and Thrissur, besides Theni, Tirunelveli, Coimbatore and Kanyakumari districts of Tamil Nadu.

The first symptom is the appearance of water-soaked brown lesions in the spear leaves of root-wilt affected palms. Gradually these spots enlarge and coalesce resulting in extensive rotting. As the leaf unfurls the rotten portions of the lamina dry and get blown off in wind, giving a 'fan' shape to the leaves. Some times, the symptom becomes very acute and the spear fails to unfurl.



Brown colour spots - enlarge resulting in rotting Fan like apperance of leaves

Management

Physical Method: Remove the rotten portions from the spear and the two adjacent leaves.

Chemical Method:

Pour fungicide solution of Hexaconazol (Contaf 5E) - 2ml or Mancozeb (Indifil M45) - 3g in 300ml water per palm to the base of spindle leaf. 2-3 rounds of spraying are sufficient in case of mild infection.

Spray crowns and leaves with 1% Bordeaux mixture or 0.5% copper oxychloride formulations or 0.4% mancozeb in January, April-May and September. While spraying, care has to be taken to spray the spindle leaf.

[Top of page](#) | [Disease Management](#)

5. GREY LEAF SPOT: *Pestalotia palmivora*

Symptoms of Damage

Minute yellow spots encircled by grayish bands appear on the surface of mature leaves of the outer whorl.

Later they become grayish white surrounded by yellow halo. These spots coalesce into irregular necrotic patches.

Complete drying and shriveling of the leaf blade are common when the infection is severe.



Minute yellow spots encircled by grayish bands

GREY LEAF SPOT: *Pestalotia palmivora*

Management

Physical Method: Removal of the older 2-3 disease affected leaves.

Chemical Method: Spraying the foliage with 0.25% Copper oxychloride will check the spread of the disease. Remove severely affected older leaves and burn. Spray the trees with 1% Bordeaux mixture or propiconazole 0.3%.

[Top of page](#) | [Disease Management](#)

6. STEM BLEEDING DISEASE: *Thielaviopsis paradoxa*

The progress of the disease is faster during July to November.

Symptoms of Damage

Stem Bleeding is characterized by the exudation of a dark reddish brown liquid from the longitudinal cracks in the bark and wounds on the stem trickling down for a distance of several inches to several feet.



Exudation of reddish-brown liquid through cracks on a coconut trunk

The lesions spread upwards as the disease progresses. The liquid oozing out dries up and turns black. The tissues below the lesions become rotten and turn yellow first and later black.

In advanced cases, the interior of affected trunks are hollow due to decay of interior tissues. As a result of extensive damage in the stem tissue, the outer whorl of the leaves turn yellow, dry and shed prematurely. The production of bunches is affected adversely. Nut fall is also noticed.

Causal Organism

The fungus, *Thielaviopsis paradoxa* is the causal agent of the disease. Progress of the disease was faster during July to November. Increase in growth cracks on the trunk, severe summer followed by sudden wetting, imbalanced nutrition, excessive salinity etc aggravates the disease. The fungus has also been found sometimes to attack leaves, petioles and nuts.

Management

Chemical Method:

Chisel out completely the affected tissues and paint the wound with tridemorph 5% or Bordeaux paste. Apply coal tar after 1-2 days on the treated portion. Burn off chiseled pieces.

To avoid spread of disease on to upper portion of trunk Root feed with tridemorph (5% Calixin) 5ml in 100 ml water, thrice a year during April-May, September-October and January-February to prevent further spread of lesions.

Cultural Method:

Avoid any mechanical injury to trunk.

Along with 50 kg FYM, apply 5kg neem cake containing the antagonistic fungi, *Trichoderma* @ 200g/palm/yearculture to the basin during September.

Provide adequate irrigation during summer and drainage during rainy season along with recommended fertilizer.

[Top of page](#) | [Disease Management](#)

7. LEAF BLIGHT (LLB) *Lasiodiplodia theobromae*

The leaf blight disease of coconut caused by the fungus *Lasiodiplodia (Botryodiplodia) theobromae* (Pat.) Griffon and Maubl. is an emerging serious problem in Pollachi tract of Tamil Nadu. At present, the disease is spreading at a faster rate in Coimbatore, Erode, Dindigul, Tirunelveli, Kanyakumari and other districts of Tamil Nadu and causing 10 – 25 per cent yield loss.

Symptoms

- Leaf blight causes serious damage in seedlings and adult palms.
- The pathogen causes damage in leaf and nuts.
- Generally the adult leaves in the lower 3 to 4 whorls are affected.
- The affected leaflets start drying from the tip downwards and exhibit a charred or burnt appearance.
- Dark grey to brown lesions with wavy to undulated margins appear from the apex of the nuts.
- The fungus entered in to the kernel through mesocarp, resulting in a decay of the endosperm.
- The affected nuts were desiccated, shrunk, deformed and dropped prematurely and resulting in nut yield loss up to 10 to 25%.
- The incidence was noticed through out the year. Maximum incidence was observed during summer months.
- Spores and the resting structures on the affected portion of the leaves served as inoculum for further spread through wind.

Control measures:

Following integrated approaches are recommended

Cultural method:

- Remove and burn the severely affected leaves to avoid further spread.

Biological method:

- Application of 200g *Pseudomonas fluorescens* along with 50 kgs of FYM+5 kgs of Neemcake/palm/yr.

Chemical method:

- Spray 1.0 per cent Bordeaux mixture or 0.25 per cent Copper oxychloride (2 times at 45 days interval during summer months).
- Root feeding of Carbendazim 2 g or Hexaconazole/ Tridemorph 2 ml + 100 ml water (3 times at 3 months interval)
- Application of an additional quantity of 2 kgs of MOP.



Yellow spots on leaves encircled by greyish on leaflets, later become grayish white

a. Preparation of 1% Bordeaux mixture

A quantity of 400 g of copper sulphate should be dissolved in 20 litres of water and 400 g of lime in another 20 litres of water separately. The copper sulphate solution should be added to the lime solution constantly stirring the mixture. Earthen or wooden vessels alone should be used and metallic containers should not be used. To find out whether the mixture is in correct proportion, a polished knife should be dipped in the mixture for one minute and taken out. If there is reddish brown deposit of copper, additional quantity of lime should be added till there is no deposit in the knife.

b. Preparation of Bordeaux paste

Take 200 g of Copper sulphate and dissolve it in one litre of water and 200 g of lime in one litre of water separately. Both are mixed simultaneously in a third vessel and the resultant mixture can be used as a paste.

[Top of page](#) | [Disease Management](#)

8. CROWN CHOCKING

Crown choking disorder is commonly observed in Assam and West Bengal.

Symptoms of Damage

Characterized by emergence of shorter leaves with fascinated and crinkled leaves.

The leaflets show severe tip necrosis and fail to unfurl. In many cases, it gives a choked appearance to the frond.

Ultimately the affected palm dies.

Control measures

Application of 50 g Borax at half-yearly intervals (Feb-Mar and Sept-Oct) along with recommended fertilizer in the basins will control the disease when it is in the early stage. In root wilt affected areas a dosage of 200gm - 300gm per palm per year is recommended.